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SAM3/ESR12 Selection Committee
Empa – Swiss Federal Laboratories for Materials Science and Technology
Dübendorf, Switzerland

Re: Application for SAM3/ESR12 PhD Position – Design, Synthesis, and Multiscale Characterization of Materials for Soft Robotics Applications

Dear SAM3 Selection Committee,

I am writing to express my strong interest in the SAM3/ESR12 PhD position on the design, synthesis, and multiscale characterization of materials for soft robotics applications at Empa and EPFL. The project's combination of functional materials, architectured structures, experimental characterization, and medical soft robotics strongly resonates with the research direction I have gradually developed throughout my Master's studies.

Although my academic background is in Computer Science, I have found that I thrive most in experimental and interdisciplinary research environments, where I can work directly with physical systems, build and test prototypes, and learn through experimentation. My interest in soft robotics began with a Soft Robotics course during my Master's programme at the University of Freiburg. What initially started as a course interest gradually developed into a research direction: I subsequently joined the LivMatS MasterLab, where I worked on a bistable pneumatic soft actuator based on flexible mechanical metamaterials, and I am now conducting my Master's thesis at the Max Planck Institute for Intelligent Systems, working on soft sensing and robotic perception.

At LivMatS, I worked on the optimization and experimental characterization of a bistable pneumatic soft actuator based on flexible mechanical metamaterials. I developed a pressure-characterization setup integrating a motorized syringe pump, pressure sensors, and Arduino-based data acquisition to measure the actuator's snap-through and snap-back behaviour and compare experimental results with theoretical predictions. This work resulted in a publication at ACTUATOR 2024 and strengthened my interest in understanding how material architecture, fabrication, and mechanical behaviour can be translated into functional robotic systems. I also gained interdisciplinary research experience during a six-month research internship at OIST in Japan.

I am currently completing my MSc in Computer Science while conducting my Master's thesis at the Max Planck Institute for Intelligent Systems in Stuttgart. My thesis, *Smart Soft Sensor Architectures for Multimodal Perception*, focuses on developing conductive hydrogel-based sensing interfaces for a sensorized endoscope tip for robotic perception. My work spans hydrogel fabrication, bio-inspired contact geometries, experimental testbed development, data acquisition, and characterization of multimodal tactile responses. Working hands-on with these materials and experimental systems has further reinforced my interest in the physical foundations of intelligent robotic systems.

What particularly attracts me to SAM3 is the opportunity to take this progression further into the design, synthesis, and multiscale characterization of functional materials for medical soft robotics. While I do not come from a traditional materials science or mechanical engineering background, I bring a strong willingness to learn and a demonstrated ability to move across disciplines and develop practical expertise through hands-on research. I am particularly eager to deepen my knowledge through the project's work on the design of programmable functional metamaterials for soft robotics, the development of 3D-printing processes for ceramic-particle-filled hydrogels, and the characterization of their morphology, composition, micromechanical behaviour, and functional properties. I am especially excited by the opportunity to understand how material composition, processing, and architecture can be connected to mechanical and functional performance. I see the opportunity to learn these aspects through hands-on research as one of the most exciting aspects of the project.

The medical focus of SAM3 is also interesting to me. My current work on a sensorized endoscope has made me increasingly interested in how soft materials, sensing, and robotic systems can be combined to enable new approaches to minimally invasive procedures. The opportunity to contribute to the development and

characterization of functional materials and structures for such applications would therefore be a natural and exciting continuation of my research journey.

I would be very excited to contribute to the SAM3 consortium and bring my background in sensing, computation, experimental prototyping, and soft robotics to the project. More importantly, I am eager to continue learning, take on the materials-focused challenges of the project, and develop into a stronger interdisciplinary researcher at the intersection of functional materials, soft robotics, and intelligent robotic systems.

Thank you very much for considering my application. I would be grateful for the opportunity to contribute to this project and to further develop my research expertise within the SAM3 network.

Kind regards,
Ritika Gupta